

Exercise 1:

- 1) Write a python program that reads the amount of time in seconds and then displays the equivalent hours, minutes and remaining seconds.

▪ Sample:

```
Enter the number of seconds: 8500
The amount of time:  2 : 21 : 40
```

Solution:

```
1. second = eval(input("Enter the number of seconds: "))
2.
3. minute = second // 60
4. second = second % 60
5. hour = minute // 60
6. minute = minute % 60
7.
8. print("The time: ", hour, ":", minute, ":", second)
9.
```

- 2) Write a python program that given a 3-digit integer number, converts it out to an integer number in reversed order.

▪ Sample:

```
Enter three digits number: 425
The number in reversed order: 524
```

Solution:

```
1. nbr = eval(input("Enter three digits number: "))
2.
3. rev = nbr % 10
4. nbr = nbr // 10
5. rev = rev * 10 + nbr % 10
6. nbr = nbr // 10
7. rev = rev * 10 + nbr % 10
8.
9. print("The number in reversed order: ", rev )
```

- 3) Write a python program that given the width and length of a garden in meters calculates the area of the garden together with the number of trees that could be planted on this garden area, knowing that each tree needs a space of 50 cm². Note: You can use the function int(x/y) that gives the integer part of the result of the division x/y.

▪ Sample:

```
Enter length of the garden in meters: 5
Enter width of the garden in meters: 6.5
The number of trees: 65 trees
```

Solution:

```
1. length = eval(input("Enter length of the garden in meters: "))
2. width = eval(input("Enter width of the garden in meters: "))
3.
4. area = length * width
5.
6. nbr_of_trees = int(area / 0.5)
7.
8. print(f"The number of trees: {nbr_of_trees} trees")
```

Exercise 1:

- 4) You want to cook some pasta, but you are short of time. To be sure you can finish cooking before your next appointment, you need to know how long it takes for the water to boil. At its highest setting, your stove needs two minutes per liter to reach the boiling point. You use a cylindrical pot. Write a program that, given the diameter of the pot and the height of the water in it, calculates the time needed for the water to boil. $\text{volume of cylinder} = \pi r^2 h$.

▪ Sample:

```
Enter diameter of the bot: 5
Enter height of the bot: 13
Your stove needs: 0.51025 minutes
```

Solution:

```
1. diameter = eval(input("Enter diameter of the bot: "))
2. height = eval(input("Enter height of the bot: "))
3.
4. PI = 3.14
5.
6. bot_volume = PI * height * (diameter / 2)** 2
7. time_to_boiling = 2 * (bot_volume / 1000)
8.
9. print("Your stove needs:", time_to_boiling, "minutes")
```

- 5) The Pythagorean Theorem states that the sum of the squares of the two sides of a right angle triangle is equal to the square of its hypotenuse. For example, 3, 4 and 5 are the sides of a right angle triangle as they form a Pythagorean Triple ($5^2 = 4^2 + 3^2$). Given 2 numbers, m and n where $m \geq n$, a Pythagorean Triple can be generated by the following formulae:

$$a = m^2 - n^2$$
$$b = 2 \times m \times n$$
$$c = \sqrt{a^2 + b^2}$$

Write a python program that reads in values for m and n and prints the values of the Pythagorean Triple generated by the formulae above.

▪ Sample:

```
Enter m number: 5
Enter n number: 4
Pythagorean Triple: 9.0 40 41.0
```

Solution:

```
1. import math
2.
3. m = eval(input("Enter m number: "))
4. n = eval(input("Enter n number: "))
5.
6. a = math.pow(m,2) - n ** 2
7. b = 2 * m * n
8. c = math.sqrt(a ** 2 + b ** 2)
9.
10. print("Pythagorean Triple: ", a, b, c)
```

Exercise 1:

- 6) Write a python program that will output the account balance each year for 3 years given the initial balance and interest rate. The interest is calculated for one year by multiplying the current account balance by the interest rate and adding this to the balance.

- Sample:

```
Enter the initial balance: 1000
Enter interest rate: 10
balance of first year = 1100.0 $
balance of second year = 1210.0 $
balance of third year = 1331.0 $
```

Solution:

```
1. balance = eval(input("Enter the initial balance: "))
2. rate = eval(input("Enter interest rate: "))
3.
4. balance = balance + (balance * rate / 100)
5. print(f"balance of first year= {balance} $")
6.
7. balance = balance + (balance * rate / 100)
8. print(f"balance of second year= {balance} $")
9.
10. balance = balance + (balance * rate / 100)
11. print(f"balance of third year= {balance} $")
```

- 7) Get the Money Write an algorithm that reads the amount of money in pennies and displays the equivalent dollars, quarters, dimes, nickles and pennies.

- One dollar corresponds to 100 pennies.
- One quarter corresponds to 25 pennies.
- One dime corresponds to 10 pennies.
- One nickle corresponds to 5 pennies.

- Sample:

```
Enter the amount of money in pennies: 843
The amount of money = 8 Dollars 1 Quarters 1 Dimes 1 Nickles 3 pennies
```

Solution:

```
1. penny = eval(input("Enter the amount of money in pennies: "))
2.
3. dollar = penny // 100
4. penny = penny % 100
5.
6. quarter = penny // 25
7. penny = penny % 25
8.
9. dime = penny // 10
10. penny = penny % 10
11.
12. nickle = penny // 5
13. penny = penny % 5
14.
15. print("The amount of money = ", dollar, "Dollars", quarter,
16. "Quarters", dime, "Dimes", nickle, "Nickles", penny, "pennies")
```

Exercise 1:

MCQ:

1) Which of the following is an invalid variable:

- 1- my_string_1
- 2- 1st_string
- 3- foo
- 4- _

Solution:

- 2- 1st_string

2) What is the answer to this expression, $22 \% 3$ is:

- 1- 7
- 2- 1
- 3- 0
- 4- 5

Solution:

- 2- 1

3) Given a function that does not return any value, What value is thrown by default when executed in shell:

- 1- int
- 2- bool
- 3- void
- 4- None

Solution:

- 4- None

4) The expression `int(x)` implies that the variable `x` is converted to integer:

- 1- True
- 2- False

Solution:

- 1- True